# 1 Model

We develop a static general equilibrium model of multinational production in which the strategic agent is the ownership group. A group can operate plants in several countries, chooses which of its plants serve each market, and takes into account when it sets quantities that its own plants compete with one another. Markups therefore depend on the group's total share of the market rather than on the share of any single plant. Profits are paid to the countries that own each group, and ownership is recorded group by group rather than country by country. Proofs are in the Appendix.

## 1.1 Model Set-Up

**Agents.** There is a finite set of countries $\mathcal{N} = \{1, \dots, N\}$ and a finite set of sectors $\mathcal{K} = \{1, \dots, K\}$. Each country $n$ is inhabited by a representative household that supplies $L_n$ units of labor inelastically and owns shares in firms. Production takes place in plants, and every plant belongs to exactly one group $g \in \mathcal{G}$, which we identify with its global ultimate parent. Each group has a headquarters country $h(g)$ and operates in a single sector $k(g)$.[1] Groups are of two types. A *parent group* $g \in \mathcal{G}^P$ is headquartered in a country of a set $\mathcal{N}^P \subseteq \mathcal{N}$ and has one potential plant in every country of $\mathcal{N}^P$, its headquarters country included. A *stand-alone firm* $g \in \mathcal{G}^S$ has a single potential plant, located in $h(g)$; countries outside $\mathcal{N}^P$ host only stand-alone firms. We index plants by $i$, write $g(i)$ and $\ell(i)$ for the group and the location of plant $i$ and $k(i) = k(g(i))$ for its sector, and let $\mathcal{P}_g$ be the set of potential plants of group $g$. Each plant produces one differentiated variety and can sell it in any country. A *market* is a destination–sector pair $(n, k)$. Whether a group is multinational is an equilibrium outcome: a group is multinational if its active plants are located in more than one country, an active plant located outside its group's headquarters country is a foreign affiliate, and the home plant of a multinational group is a domestic multinational. A stand-alone firm is never multinational.

**Representative household.** The household in country $n$ has nested CES preferences,

$$U_n = \Big[\sum_{k \in \mathcal{K}} \beta_k^{1/\eta}\, C_{nk}^{(\eta-1)/\eta}\Big]^{\eta/(\eta-1)}, \qquad C_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} q_{i,n}^{(\sigma_k-1)/\sigma_k}\Big]^{\sigma_k/(\sigma_k-1)}, \tag{1}$$

where $\mathcal{A}_{nk}$ is the set of plants active in market $(n, k)$, $\beta_k > 0$ with $\sum_k \beta_k = 1$, and $\sigma_k > \eta \geq 1$: varieties within a sector are closer substitutes than goods in different sectors. The case $\eta = 1$ is the Cobb–Douglas limit. The associated price indices are

$$P_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} p_{i,n}^{1-\sigma_k}\Big]^{1/(1-\sigma_k)}, \qquad P_n = \Big[\sum_k \beta_k P_{nk}^{1-\eta}\Big]^{1/(1-\eta)}. \tag{2}$$

The household spends its income $X_n$, allocates the share $\beta_k P_{nk}^{1-\eta}/P_n^{1-\eta}$ of it to sector $k$, and its welfare is real income $X_n/P_n$. Firms buy intermediate inputs as the same sectoral composites (below), so all purchases of sector-$k$ varieties in country $n$, by households and by firms, are aggregated by $C_{nk}$. Writing $q_{i,n}$ for the total quantity of plant $i$'s variety sold in $n$ and $E_{nk}$ for total spending on sector $k$ in $n$, demand is

$$q_{i,n} = p_{i,n}^{-\sigma_k}\, P_{nk}^{\sigma_k - 1} E_{nk}. \tag{3}$$

[1] This restriction is without loss for behavior. Marginal costs are constant and fixed costs are specific to each plant–market pair, so a group's decisions separate across markets (Section 1.2). A group active in several sectors is equivalent to several single-sector groups with the same owners; what the restriction rules out is a capability shared across sectors.

Market sizes $E_{nk}$ are determined in general equilibrium (Section 1.3).[2]

**Production technology.** Labor is the only primary factor. A plant in sector $k$ located in $\ell$ combines labor with a bundle of intermediate inputs. Writing $h_i \equiv h(g(i))$ for the headquarters country of plant $i$, its unit cost of production is

$$c_i = \frac{1}{\varphi_i}\, w_\ell^{1-\nu_k} \left(P_{\ell k}^{\mathrm{IO}}\right)^{\nu_k} \gamma_{h_i \ell}, \qquad P_{\ell k}^{\mathrm{IO}} = \prod_{k' \in \mathcal{K}} P_{\ell k'}^{\omega_{k'k}}, \tag{4}$$

where $w_\ell$ is the wage, $\nu_k \in [0,1)$ is the share of intermediates in costs, and $\omega_{k'k} \geq 0$, with $\sum_{k'} \omega_{k'k} = 1$, is the share of sector $k$'s intermediate spending that falls on sector $k'$ (Caliendo and Parro, 2015); the constant of the Cobb–Douglas cost function is absorbed into $\varphi_i$. The term $\gamma_{h\ell} \geq 1$, with $\gamma_{hh} = 1$, is the efficiency loss from producing away from headquarters (Ramondo and Rodríguez-Clare, 2013; Head and Mayer, 2019). Intermediates are bought in the markets $(\ell, k')$ of the country where the plant operates, which include imported varieties and are subject to that country's tariffs. At given wages, the entry of a plant into a market lowers its price index and hence the input costs of every producer located in that country; in general equilibrium the same entry also raises local labor demand, and the net effect on costs depends on both.

**Head-office capability.** The productivity of plant $i$ combines a location, a plant and a group component,

$$\varphi_i = z_\ell\, e^{u_i} \times \begin{cases} \exp\left[(1+\delta^P \alpha_k)\, x_g\right] & \text{if } g \in \mathcal{G}^P, \\ \exp\left[-\delta^S \alpha_k\right] & \text{if } g \in \mathcal{G}^S,\ h(g) \in \mathcal{N}^P, \\ 1 & \text{if } g \in \mathcal{G}^S,\ h(g) \notin \mathcal{N}^P, \end{cases} \tag{5}$$

where $z_\ell > 0$ is the productivity of location $\ell$, common to all plants there whatever their owner; $u_i$ is an idiosyncratic plant draw, normal with mean zero and independent across plants; $x_g \geq 0$ is the capability of parent group $g$, with $e^{x_g}$ Pareto with unit lower bound and shape $\vartheta$; and $\alpha_k \in [0,1]$ indexes the head-office intensity, or complexity, of sector $k$ (Antràs, 2003; Antràs and Helpman, 2004). Head-office services (management, technology, brands) are a non-rival capability of the group that all of its plants share, so they enter productivity rather than the cost of inputs. A stand-alone firm has no access to them, and its productivity falls short by a factor that grows with complexity at rate $\delta^S \geq 0$. A parent group's capability is worth more in complex sectors: in sector $k$ the group component $e^{(1+\delta^P \alpha_k) x_g}$ is Pareto with shape $\vartheta/(1+\delta^P \alpha_k)$, so the distribution of capability has a thicker tail where complexity is higher. Stand-alone firms in countries outside $\mathcal{N}^P$ stand in for all supply from those countries, multinational or not, and carry no penalty. The complexity index $\alpha_k$ appears only in (5).[3]

[2] Intermediate inputs are combined across sectors with Cobb–Douglas weights, so the upper-tier elasticity of intermediate demand is one. When intermediates are used ($\nu_k > 0$ below) we therefore set $\eta = 1$, so that final and intermediate demand share the same upper-tier elasticity and the markup in (10) is the one firms face. With $\eta = 1$, spending on each sector is a fixed share of total spending, so a market does not expand when its price index falls; the case $\eta > 1$ is used only without intermediates.

[3] The common alternative treats head-office services as a factor input bought at the parent's wage, so that the labor term in (4) becomes $(w_h^{\alpha_k} w_\ell^{1-\alpha_k})^{1-\nu_k}$ and the fixed cost (7) is paid in the same mix of labor (Antràs, 2003). A plant abroad then pays a foreign wage, which links factor markets across countries and can overturn the substitutability of labor across countries on which uniqueness of wages rests (Appendix C).

**Pool of potential groups.** The set of potential groups is drawn once, before any decision is taken, and the equilibrium is computed for a given draw. Each country–sector pair $(n, k)$ has a given number $M^S_{nk}$ of potential stand-alone firms, and each sector $k$ has a given number $M^P_k$ of potential parent groups. The headquarters country of each potential parent group is drawn independently from $\mathcal{N}^P$ with probability proportional to $L_h z_h^\zeta$, with $\zeta \geq 0$, and its capability $x_g$ is drawn from the same distribution whatever its headquarters. Larger and more productive countries therefore host more potential parents, and their best parents tend to be better only because more draws are taken there (Gaubert and Itskhoki, 2021). Because each market contains a finite number of groups, the draw of the pool matters for outcomes, and every equilibrium object below is conditional on it.

**Trade.** Shipping goods from $\ell$ to $n$ entails an iceberg cost $d_{\ell n} \geq 1$, with $d_{nn} = 1$. Trade costs are incurred from the country where the good is produced, not from headquarters, so a single plant can serve several destinations, including third markets, as an export platform (Tintelnot, 2017). Country $n$ levies an ad valorem tariff $t_{\ell nk} \geq 0$ on sector-$k$ goods produced in $\ell$, with $t_{nnk} = 0$. The delivered marginal cost of plant $i$ in market $(n, k)$ is

$$c_{i,n} = c_i \, d_{\ell n} \, (1 + t_{\ell nk}), \qquad \ell = \ell(i). \tag{6}$$

The tariff is levied on the value of shipments at marginal cost, so country $n$ collects $t_{\ell nk} \, c_i d_{\ell n} q_{i,n}$ on the sales of plant $i$.[4]

**Multinational corporations and entry.** A plant that serves market $(n, k)$ pays a fixed cost

$$F_{i,n} = f_k \, w_\ell \, f^d_{\ell n}, \tag{7}$$

in units of labor of the country where the plant is located, with $f_k > 0$ and $f^d_{\ell n} \geq 1$ raising the cost of serving distant markets. The cost is incurred once per market served, whatever the quantity sold, and stands for registration, distribution and marketing. Because it is paid in labor, entry uses real resources, appears as wage income, and leaves the model homogeneous of degree one in wages. Marginal costs are constant and fixed costs are specific to each plant–market pair, so a group's problem separates across destinations: in each market $(n, k(g))$ the group chooses the subset $T_{g,n} \subseteq \mathcal{P}_g$ of its plants that serve $n$. A parent group can serve a market from a plant at home, from a plant in the destination, from a plant in a third country, or from several at once; a stand-alone firm decides only whether to serve $n$. A plant operates if it serves at least one market. Which affiliates exist, and where each of them sells, is thus an equilibrium outcome, in the spirit of the multi-plant location problem of Yang (2023).

**Profit repatriation.** Country $n$ owns a share $\theta_{gn} \geq 0$ of group $g$, with $\sum_n \theta_{gn} = 1$. The profit of the group net of fixed costs, $\Pi_g = \sum_n \big[\Pi_{g,n} - \sum_{i \in T_{g,n}} F_{i,n}\big]$, where $\Pi_{g,n}$ is its variable profit in market $(n, k(g))$ (Section 1.2), is paid to its owners in proportion to $\theta_{gn}$, wherever it is earned. Income in country $n$ is

$$X_n = w_n L_n + \sum_{g \in \mathcal{G}} \theta_{gn} \, \Pi_g + T_n, \qquad T_n = \sum_k \sum_{i \in \mathcal{A}_{nk}} t_{\ell(i)nk} \, c_i \, d_{\ell(i)n} \, q_{i,n}. \tag{8}$$

[4] Because markups are multiplicative, the price paid in $n$ is the same as under a duty levied on the exporter's price $p_{i,n}/(1 + t_{\ell nk})$. The two specifications differ only in how the wedge $t_{\ell nk}(\mu - 1)c_i d_{\ell n} q_{i,n}$ is divided between the treasury of $n$ and the group.

Profits are earned where groups sell and received where their owners reside, and $\theta$ is what separates the two. A natural benchmark assigns each group to the country of its ultimate parent, $\theta_{gn} = \mathbf{1}\{n = h(g)\}$, which is what parent identity in firm-level data identifies; the results below hold for any ownership matrix.[5]

**Timing.** The model is static. First, the pool of potential groups and all productivity draws are realized. Second, given wages, price indices and market sizes, each group chooses which of its plants serve each market of its sector. Third, active plants compete market by market, production and trade take place, and fixed costs are paid. Fourth, profits are distributed to owners and households spend their income. In equilibrium all choices are mutually consistent; the order is expository and matches the solution algorithm of Section 1.3.

## 1.2 Market Structure

Fix a market $(n, k)$ and suppress its indices, writing $\sigma$, $E$ and $P$ for $\sigma_k$, $E_{nk}$ and $P_{nk}$, $c_i$ for the delivered cost $c_{i,n}$, and $T_g$ for the plants of $g$ active in the market.

**Demand.** Let $A \equiv \sum_{i \in \mathcal{A}} p_i^{1-\sigma} = P^{1-\sigma}$. By (3), the revenue share of plant $i$ and the share of group $g$ are

$$s_i = \frac{p_i q_i}{E} = \frac{p_i^{1-\sigma}}{A}, \qquad S_g = \sum_{i \in T_g} s_i, \qquad \sum_g S_g = 1. \tag{9}$$

The aggregate $A$ measures how competitive the market is: when rivals charge low prices, $A$ is large and a given price buys a small share.

**Competition.** Groups are the strategic agents. Each group chooses the quantities of all of its plants serving the market, taking as given the quantities of other groups and the level of spending, and maximizes variable profit $\Pi_g = \sum_{i \in T_g} (p_i - c_i) q_i$.

**Proposition 1** (Markups)**.** *Under quantity competition, the first-order conditions of group $g$ are $p_i = \mu(S_g)\, c_i$ for every $i \in T_g$, with*

$$\frac{1}{\mu(S)} = 1 - \frac{1 - S}{\sigma} - \frac{S}{\eta}. \tag{10}$$

*Under price competition the markup is $\mu(S) = \varepsilon(S)/[\varepsilon(S) - 1]$ with $\varepsilon(S) = \sigma - (\sigma - \eta)S$.*

Both are the markups of Atkeson and Burstein (2008), evaluated at the share of the group rather than of the plant. All results below rely only on the following property, which both satisfy whenever $\sigma > \eta$.

**Assumption 1.** $\mu$ is continuously differentiable and strictly increasing on $[0, 1)$, with $\mu(0) = \sigma/(\sigma - 1)$.

We take quantity competition as the baseline and report where conclusions depend on it.[6]

[5] Income includes the profits of the groups a country owns wherever they produce and sell, so trade need not balance country by country. World income equals world final spending, which follows from (8), (16) and (17) by Walras' law.

[6] When $\eta = 1$ the markup in (10) is unbounded as $S \to 1$, so a market served by a single group has no equilibrium. We restrict attention to configurations in which at least two groups are active in every market.

**Cannibalization across multinational affiliates.** When a group expands one of its plants, the sector price index falls and every variety in the market loses sales, the group's other plants included. A single-plant firm ignores the losses it imposes on rivals; a group counts the losses it imposes on itself. On the fraction $1 - S_g$ of the market held by rivals, the group faces the within-sector elasticity $\sigma$. On the fraction $S_g$ it already holds, an expansion of one plant mostly displaces another of its own plants, and the relevant elasticity is the across-sector elasticity $\eta$. Equation (10) weights the two, so the markup rises with the group's share, from $\sigma/(\sigma-1)$ for a negligible group toward the monopoly markup. Two implications follow. First, all plants of a group charge the same markup in a given market, and differences in their costs show up in quantities rather than in markups. Second, market power depends on the total share of the group, so counting plants under their common owner raises both measured concentration and markups. Within a group, (9) and the common markup give

$$\frac{s_i}{S_g} = \frac{c_i^{1-\sigma}}{K_g}, \qquad K_g \equiv \sum_{i \in T_g} c_i^{1-\sigma}. \tag{11}$$

We call $K_g$ the *strength* of the group in the market: an index of its cost efficiency that rises with the number of plants serving the market and with how cheaply each delivers. Substituting $p_i = \mu(S_g)c_i$ into (9),

$$\psi(S_g) = \frac{K_g}{A}, \qquad \psi(S) \equiv S\,\mu(S)^{\sigma-1}. \tag{12}$$

Under Assumption 1 the map $\psi$ is strictly increasing, so $S_g = \psi^{-1}(K_g/A)$: the share of a group depends only on its strength relative to the competitiveness of the market. However many plants a group runs, and in however many countries, its position in the market is summarized by one number, $K_g$, and all it needs to know about its rivals is $A$. The equilibrium of the market is the value of $A$ at which shares add up to one,

$$\sum_g \psi^{-1}\left(\frac{K_g}{A}\right) = 1. \tag{13}$$

**Profits.** Variable profit is $\Pi_g = E\,S_g\,m(S_g)$, where $m(S) \equiv 1 - 1/\mu(S)$ is the Lerner index. Under quantity competition (10) makes the Lerner index linear in the share, $m(S) = (1 + \kappa S)/\sigma$ with $\kappa \equiv \sigma/\eta - 1$, so that

$$\Pi_g = \frac{E}{\sigma}(S_g + \kappa S_g^2). \tag{14}$$

Under price competition the Lerner index $1/\varepsilon(S)$ is not linear, and $\kappa = (\sigma - \eta)/\sigma$ is the slope of its first-order expansion. Profits rise more than proportionally with the share, because a larger group both sells more and earns more on each unit sold. The curvature $\kappa$ sets the size of every ownership result in Section 1.4; with constant markups $\kappa = 0$.

**Entry.** A group's plant set matters for its profit only through its strength. Let $K(T) = \sum_{i \in T} c_i^{1-\sigma}$ and $\Phi(T) = \sum_{i \in T} F_{i,n}$ be the strength and the fixed cost of a set $T \subseteq \mathcal{P}_g$. Given $A$, serving the market with $T$ yields

$$V_g(T; A) = \Omega(K(T); A) - \Phi(T), \qquad \Omega(K; A) \equiv E\,\psi^{-1}(K/A)\,m(\psi^{-1}(K/A)). \tag{15}$$

A group with $|\mathcal{P}_g|$ potential plants chooses among $2^{|\mathcal{P}_g|}$ sets. Because $\Omega$ is strictly increasing in $K$, a set can be optimal only if no other set delivers at least as much strength at no greater fixed cost. These efficient sets form a chain, ordered by strength and by cost together, so the group's choice

amounts to how far to climb a ladder that runs from small and cheap to large and expensive. The entry decision is a cutoff, as in Gaubert and Itskhoki (2021), but a cutoff on the group's own list of plants rather than on a ranking of firms. A stand-alone firm has a ladder with a single rung.

**Definition 1** (Entry equilibrium in a market)**.** Given delivered costs $\{c_i\}$, fixed costs $\{F_{i,n}\}$ and spending $E$, an entry equilibrium is a collection of active sets $\{T_g\}$ and an aggregate $A$ such that (i) each $T_g$ maximizes $V_g(\cdot\,; A)$ over subsets of $\mathcal{P}_g$, and (ii) $A$ solves (13) with $K_g = K(T_g)$.

Groups take the competitiveness of the market as given when they decide which plants to operate, although they internalize their effect on $A$ when they set quantities. This is the notion implicit in a free-entry condition: the entry margin is evaluated at the prevailing degree of competition. Appendix B discusses the stronger notion in which every deviation is evaluated on the re-solved market.

### 1.3 General Equilibrium

**Market sizes.** Let $V_{i,m} \equiv c_i\, d_{\ell(i)m}\, q_{i,m}$ be the variable cost of plant $i$ in serving market $m$, exclusive of tariffs. By Proposition 1, $V_{i,m} = r_{i,m}/[\mu\,(1+t)]$, where $r_{i,m}$ is the revenue of the plant in that market. Spending on market $(n,k)$ is final spending by households plus intermediate purchases by every plant located in $n$, wherever it sells,

$$E_{nk} = \frac{\beta_k P_{nk}^{1-\eta}}{P_n^{1-\eta}}\, X_n + \sum_{k' \in \mathcal{K}} \omega_{kk'}\, \nu_{k'} \sum_{m \in \mathcal{N}} \sum_{i \in \mathcal{A}_{mk'}:\, \ell(i)=n} V_{i,m}. \tag{16}$$

**Labor market.** The wage bill of country $n$ pays for the labor used in production and the labor used to serve markets,

$$w_n L_n = \sum_{k \in \mathcal{K}} (1-\nu_k) \sum_{m \in \mathcal{N}} \sum_{i \in \mathcal{A}_{mk}:\, \ell(i)=n} V_{i,m} \;+\; \sum_{m \in \mathcal{N}} \sum_{i \in \mathcal{A}_{mk(i)}:\, \ell(i)=n} F_{i,m}. \tag{17}$$

**Definition 2** (Equilibrium)**.** An equilibrium consists of wages $\{w_n\}$, price indices $\{P_{nk}\}$, market sizes $\{E_{nk}\}$, incomes $\{X_n\}$, active sets $\{T_{g,n}\}$, and prices and quantities $\{p_{i,n}, q_{i,n}\}$ such that

(i) in every market, each group's quantities maximize its variable profit given its rivals', so that prices satisfy Proposition 1;

(ii) in every market, quantities satisfy demand (3) and shares satisfy (13);

(iii) in every market, the active sets and the aggregate form an entry equilibrium (Definition 1), with fixed costs (7);

(iv) costs (4)–(6) are evaluated at the price indices (2) that they generate;

(v) market sizes satisfy (16) and incomes satisfy (8);

(vi) labor markets clear, (17).

All prices, costs and incomes are homogeneous of degree one in wages and all shares are homogeneous of degree zero, so wages are determined up to scale. By Walras' law one labor market condition is implied by the others, and we set $w_1 = 1$.

**Existence and uniqueness.** Given wages, the model solves as a sequence of nested problems, and we establish existence and uniqueness layer by layer. The first two layers are single markets.

**Proposition 2** (One market)**.** *Under Assumption 1, for any active set containing at least two groups and any positive costs, the market has exactly one equilibrium.*

The proof uses (13): each $\psi^{-1}(K_g/A)$ falls as $A$ rises, so the sum of shares is a strictly decreasing function of a single number and equals one exactly once. Each group's profit is strictly concave in its own quantities, so the first-order conditions identify the unique best response.

Entry is harder. Each entry raises $A$ and lowers the value of entering for everyone else, and discrete games of this kind typically have several equilibria. The usual remedy, ranking entrants by cost and letting the most efficient enter first, requires that every entrant be a separate firm; once a group can add one of its own plants, an entry raises its share instead of lowering it and the ranking argument fails. Fixing an order of moves restores uniqueness only by selecting among equilibria. We instead separate the effect of a group's strength on its own payoff from the effect of its rivals. Write $\Omega_K$ and $\Omega_{KA}$ for the partial derivatives of $\Omega$.

**Theorem 1** (Entry)**.** *Consider quantity competition with $\eta = 1$ and $\sigma \geq 2$.*

*(a) $\Omega(K; A)$ is strictly concave in $K$ at every share.*

*(b) $\Omega_{KA} < 0$ whenever the group's share is at most one half.*

*(c) Suppose that no group's share exceeds one half at any strength profile between two candidate equilibria, and that no group is indifferent between two rungs of its ladder. Then a market has at most one entry equilibrium.*

Part (a) says that strength has diminishing returns for the group itself, and part (b) that strength is worth less in a more competitive market. Together they imply that each group's optimal strength falls as $A$ rises, while $A$ rises with every group's strength, so two equilibria with different aggregates cannot both exist. No order of entry is imposed and groups continue to internalize competition among their own plants. The hypothesis has economic content and can be checked: no single group holds more than half of a market. It is sufficient rather than necessary; at exactly one half, the expression whose sign gives (b) still equals $1/\sigma$. The proof covers quantity competition with $\eta = 1$. For price competition and for $\eta > 1$ we have verified numerically that the same one-half condition suffices, but we do not have a proof. Existence is a separate matter: entry is chosen in whole plants, so an exact equilibrium in pure strategies need not exist (Appendix B).

The remaining layers take active sets as given.

**Proposition 3** (Prices and incomes)**.** *Fix wages and active sets. (a) The price indices are the unique fixed point of* (2) *combined with* (4)–(6)*, and iteration converges to it at rate* $\max_k \nu_k$*. (b) Market sizes and incomes are the unique solution of the linear system* (8), (16)*.*

**Proposition 4** (Existence of wages)**.** *Fix active sets and suppose every country hosts at least one active plant. Then there exist wages that clear every labor market.*

Uniqueness of wages is the last layer and the most delicate. Let $W_n(w)$ be the wage bill of country $n$ implied by (17) at wages $w$, and $M_{nj} \equiv \partial \ln W_n / \partial \ln w_j$. Homogeneity makes every row of $M$ sum to one. Labor in different countries is a gross substitute when $M_{nj} > 0$ for $n \neq j$: a rise in any other country's wage raises country $n$'s wage bill. The next lemma is the building block, and it holds without conditions.

**Lemma 1** (Cost pass-through)**.** *Fix active sets. The matrix $\partial \ln P / \partial \ln w$, with rows indexed by markets, is non-negative and its rows sum to one.*

A rise in one country's wage raises every price index in the world by a proportion between zero and one, whatever the markups and the input–output structure. At each step of the argument the response is a weighted average of responses one step down: a plant's cost averages wages and input prices, and a market's price index averages the cost changes of the groups in it, with weights that variable markups change but cannot make negative.

The model's solution algorithm adjusts wages by the damped tâtonnement

$$\Psi(w)_n = w_n^{1-\lambda}\big(W_n(w)/L_n\big)^{\lambda}, \qquad \lambda \in (0,1),$$

which raises a country's wage when its labor is in excess demand and whose fixed points, up to scale, are the equilibrium wage vectors. Let $d_H(w,w') = \max_n \ln(w_n/w'_n) - \min_n \ln(w_n/w'_n)$ denote Hilbert's projective metric, which is unchanged when either vector is rescaled.

**Theorem 2** (Wages)**.** *Fix active sets. Let $D$ be a set of wage vectors that is convex in logarithms, and suppose that on $D$ (i) $M_{nj} > 0$ for all $n \neq j$, and (ii) $\lambda < 1/(1 - \min_n M_{nn})$. Then there is $\tau < 1$ such that $d_H(\Psi(w), \Psi(w')) \leq \tau\, d_H(w,w')$ for all $w, w' \in D$. Hence $D$ contains at most one equilibrium wage vector up to scale, and if $\Psi$ maps $D$ into itself the algorithm converges to it from any starting point in $D$.*

Condition (ii) is not a numerical convenience. Labor demand is elastic, so $M_{nn}$ is negative and the undamped map ($\lambda = 1$) is not order-preserving; the model itself bounds the admissible step. Condition (i) is the substantive one, and the next result gives conditions under which it holds.

**Proposition 5** (Gross substitutes)**.** *Suppose $\nu_k = 0$ and $\eta = 1$, and that no group's share exceeds one half in any market under quantity competition. Then for $n \neq j$,*

$$M_{nj} \;\geq\; \bar{E}_n^{(j)} \equiv \sum_{i,m} \pi_{i,m}^{n}\, \frac{\partial \ln E_{mk(i)}}{\partial \ln w_j}, \tag{18}$$

*where $\pi_{i,m}^{n}$ is the share of country $n$'s wage bill paid by plant $i$ out of its variable costs in market $m$. Labor is a gross substitute whenever $\bar{E}_n^{(j)} > 0$.*

The bound says that when production uses only local labor, a rise in a foreign wage can lower country $n$'s wage bill only by shrinking the markets in which its plants sell; every reallocation of sales between plants and between groups favors country $n$. The one-half condition is the same as in Theorem 1: it is what signs the effect of variable markups on factor demand. With intermediates, the same bound holds provided a plant's input prices do not rise by more than its group's average cost, a condition that is vacuous when $\nu_k = 0$.

Three limits of these results should be stated plainly. First, each layer is unique given the layers it takes as given, which does not make the joint equilibrium of wages and active sets unique: active sets change discretely with wages, and Theorem 2 holds them fixed. Second, an exact entry equilibrium in whole plants may fail to exist. Third, gross substitutes is not a theorem of the model. When head-office services are a factor input it can fail at an equilibrium (Appendix C), and with capability alone the demand term in (18) can still turn negative, because a country whose groups lose profits when a foreign wage rises spends less. That income effect is the ownership channel the model is built to study, and removing it would remove the object of interest. The conditions of Theorems 1 and 2 and of Proposition 5 are inequalities in objects the model computes, and we check them at the computed equilibrium in the quantification.

**Solution algorithm.** The structure above maps into nested loops. Given an aggregate $A$, each group's share solves the scalar equation (12), and the market solves (13) by safeguarded Newton steps on $\ln A$, which never leave a bracket that contains the unique root. With entry, each group's best response at a given $A$ is found by scanning its ladder of efficient sets, and $A$ is again found by bisection. Given wages and active sets, price indices are obtained by iterating on (4)–(2), which is a contraction, and market sizes and incomes by one linear solve. Wages are updated by the tâtonnement of Theorem 2, followed by Newton steps once close to the solution. An outer loop updates active sets: at the equilibrium wages for the current configuration, the groups with the largest gains from changing their sets do so, and the procedure repeats until no group wishes to move. When no exact fixed point in whole plants exists, the algorithm returns the configuration with the fewest groups wishing to move and reports how many do. Moving only the groups with the largest gains changes the path of the algorithm, not its fixed points, and does not amount to an order of entry.

### 1.4 Comparative Statics

**Incidence within a market.** Consider a market with given active sets and spending, and a change in delivered costs. Let $\hat{c}_g = \sum_{i\in T_g}(s_i/S_g)\, d\ln c_i$ be the average cost change of group $g$, $\varepsilon_\mu(S) \equiv S\mu'(S)/\mu(S)$ the elasticity of the markup, and

$$\Lambda_g \equiv \frac{1}{1+(\sigma-1)\,\varepsilon_\mu(S_g)}, \qquad \varpi_g \equiv \frac{S_g\Lambda_g}{\sum_h S_h\Lambda_h}. \tag{19}$$

Under quantity competition with $\eta = 1$, $\Lambda_g = (1-S_g)/[1+(\sigma-2)S_g]$, which falls from one toward zero as the share rises.

**Proposition 6** (Incidence)**.** *Holding active sets and spending fixed,*

$$d\ln S_g = -(\sigma-1)\Lambda_g(\hat{c}_g - d\ln P), \qquad d\ln P = \sum_g \varpi_g\,\hat{c}_g. \tag{20}$$

*If only group $j$'s costs change, its price rises by the fraction $\rho_j = 1-(\sigma-1)\varepsilon_\mu(S_j)\Lambda_j(1-\varpi_j) \in (0,1]$ of the cost increase, and every rival $h$ gains share, $d\ln S_h = (\sigma-1)\Lambda_h\varpi_j\hat{c}_j$, and raises its price.*

Three features distinguish this from monopolistic competition, where $\Lambda_g = 1$, $\varpi_g = S_g$ and $\rho_j = 1$. Large groups absorb part of a cost increase in their markup, so a tariff on them is partly paid by the group: pass-through $\rho_j$ falls with the group's share. Since $\Lambda_g$ falls with size, $\varpi_g < S_g$ for the larger groups in a market, and the price index moves by less than their share would suggest. And rivals are not passive: a tariff on one group raises the shares, markups and prices of the others.

**Ownership and the level of profit income.** Consider a country $H$ with ownership shares $\theta_g \equiv \theta_{gH}$, and its profit income from a market, $\Pi^H = \sum_g \theta_g\Pi_g$.

**Proposition 7** (Ownership-weighted concentration)**.** *Under quantity competition,*

$$\frac{\Pi^H}{E} = \frac{1}{\sigma}\Big[\bar{\theta} + \kappa\big(\bar{\theta}\cdot\mathrm{HHI} + \mathrm{Cov}_S(\theta,S)\big)\Big], \tag{21}$$

*where $\bar{\theta} = \sum_g S_g\theta_g$ is the sales-weighted ownership share,* $\mathrm{HHI} = \sum_g S_g^2$*, and* $\mathrm{Cov}_S(\theta,S) = \sum_g \theta_g S_g^2 - \bar{\theta}\cdot\mathrm{HHI}$*. A calculation that applies the aggregate ownership share to total profit, $\bar{\theta}\sum_g\Pi_g$, misses exactly $(E\kappa/\sigma)\,\mathrm{Cov}_S(\theta,S)$.*

The first term is what a model with constant markups delivers: ownership times revenue over $\sigma$. The second is an ownership-weighted Herfindahl index. Its component $\bar{\theta} \cdot \text{HHI}$ can be built from an aggregate ownership share and a published concentration index. The covariance between ownership and size cannot: it requires knowing which groups a country owns and how large each one is. Because profits are convex in shares, what matters for a country's profit income is not what fraction of firms it owns but what fraction of the large ones. The decomposition is an identity. It is exact under quantity competition, where the Lerner index is linear; under price competition it is the leading term of an expansion.

**Corollary 1** (Constant markups)**.** *If $\kappa = 0$, a country's profit income from a market equals its aggregate ownership share times total profit, whatever the distribution of ownership across groups.*

The corollary concerns the level of income given $\bar{\theta}$. It does not say that the distribution of ownership is irrelevant for how income responds to policy, which is the subject of the next result.

**Ownership and the response of profit income.** Policy acts through changes in profit income, not through its level, and the two are governed by different statistics.

**Proposition 8** (Response of owned profits)**.** *Under quantity competition with $\eta = 1$, holding active sets and spending fixed, a change in the delivered costs of group $j$ alone changes country $H$'s profit income by*

$$\frac{d\Pi^H}{E} = \frac{\sigma - 1}{\sigma}\, \varpi_j\, \hat{c}_j \sum_{h \neq j} S_h \Lambda_h \Big[\theta_h(1 + 2\kappa S_h) - \theta_j(1 + 2\kappa S_j)\Big]. \tag{22}$$

A cost increase on group $j$, such as a tariff on the plants that serve the market, reallocates sales from $j$ to its rivals in proportion to $S_h\Lambda_h$, and each unit of share is worth $\theta(1 + 2\kappa S)$ to country $H$. The owner gains when the rivals it owns are worth more to it at the margin than the group being taxed. Two differences from Proposition 7 matter. The weights are $S_h\Lambda_h$ rather than $S_h^2$, and $\Lambda_h$ falls with size, so large groups count for less in the response than in the level. And the response depends on the ownership of the taxed group relative to its rivals, not on the covariance of ownership with size. The ownership-weighted Herfindahl index measures the stake a country has in a market; it is not, by itself, a sufficient statistic for how that stake responds to a tariff, and in general equilibrium the response also runs through wages, entry and market sizes.

**Entry.** The proof of Theorem 1 delivers monotone comparative statics for each group's choice.

**Corollary 2** (Entry responses)**.** *Under the conditions of Theorem 1, the strength a group chooses is non-increasing in the aggregate $A$ and non-decreasing in spending $E$, holding the other fixed.*

A more competitive market leads each group to climb fewer rungs of its ladder, and a larger market to climb more. Since every rung is a set of plants, these responses determine which affiliates operate, and through (16) and (17) they feed back into market sizes and wages.

**Wages and prices.** Lemma 1 is also a comparative statics result. A rise in country $j$'s wage raises every price index, in every country and sector, by a proportion between zero and one, and an equal proportional rise in all wages raises every price index in the same proportion. A wage change therefore propagates through the input–output structure and through markups, but neither can reverse its sign or amplify it beyond one for one. How the wage change is shared between consumers in different countries depends on where the plants that use country $j$'s labor sell, and how it is shared between workers and owners depends on who owns the groups that operate those plants.

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# Appendix

## A The Market Game

**A.1 Markups.** Fix a market. With $\eta$ the upper-tier elasticity, write spending as $E = D\,P^{1-\eta}$, with the shifter $D$ taken as given by firms; $\eta = 1$ gives $E = D$. Let $\rho = (\sigma-1)/\sigma$ and $Q = (\sum_i q_i^\rho)^{1/\rho}$. Inverse demand is $p_i = D^{1/\eta} Q^{1/\sigma - 1/\eta} q_i^{-1/\sigma}$, so group $g$'s revenue is $R_g = D^{1/\eta} Q^{1/\sigma-1/\eta} \sum_{j\in T_g} q_j^\rho$. Using $\partial \ln Q/\partial q_i = q_i^{\rho-1}/Q^\rho$ and $s_i = q_i^\rho/Q^\rho$,

$$\frac{\partial R_g}{\partial q_i} = p_i\Big[\rho + \Big(\frac{1}{\sigma} - \frac{1}{\eta}\Big) S_g\Big] = p_i\Big[1 - \frac{1-S_g}{\sigma} - \frac{S_g}{\eta}\Big].$$

Setting marginal revenue equal to $c_i$ gives (10); the bracket is common to all $i \in T_g$ and depends only on $S_g$. Under price competition, demand is $q_i = p_i^{-\sigma} P^{\sigma-\eta} D$, so $\partial \ln q_j/\partial \ln p_i = -\sigma \mathbf{1}\{i = j\} + (\sigma-\eta) s_i$. The first-order condition for $p_i$ is $q_i - \sigma(p_i - c_i) q_i/p_i + (\sigma-\eta)(s_i/p_i)\sum_{j\in T_g}(p_j - c_j) q_j = 0$. Trying $p_j = \mu c_j$ for all $j \in T_g$, so that $(p_j - c_j) q_j = (1 - 1/\mu) r_j$ and $\sum_{j\in T_g} r_j = S_g E$, it reduces to $(1 - 1/\mu)\big[\sigma - (\sigma-\eta) S_g\big] = 1$, which is the stated markup.

**A.2 Proof of Proposition 2.** *The share map.* $\psi(S) = S\mu(S)^{\sigma-1}$ is the product of $S$ and of $\mu(S)^{\sigma-1}$, both positive and strictly increasing on $(0,1)$ under Assumption 1, and $\psi(0) = 0$; so $\psi$ has a continuous, strictly increasing inverse. *Best responses.* Given rivals' quantities, group $g$'s profit is $D^{1/\eta} X_g (X_g + B_g)^e - \sum_{i\in T_g} c_i q_i$, where $X_g = \sum_{i\in T_g} q_i^\rho$, $B_g$ is the same index for rivals, and $e = (\eta-\sigma)/[\eta(\sigma-1)] \in [-1, 0)$. The function $X \mapsto X(X+B)^e$ is strictly increasing and strictly concave for $B > 0$: its derivatives are $(X+B)^{e-1}[(1+e)X + B] > 0$ and $e(X+B)^{e-2}[(1+e)X + 2B] < 0$. The index $X_g$ is strictly concave in $(q_i)_{i\in T_g}$. A strictly increasing, strictly concave function of a strictly concave function is strictly concave, and cost is linear, so profit is strictly concave in the group's own quantities. Since $\partial X_g/\partial q_i \to \infty$ as $q_i \to 0$, the optimum is interior and is the unique solution of the first-order conditions. *The aggregate.* By Proposition 1 and (12), an equilibrium is an $A$ solving (13), and conversely any such $A$ generates prices $p_i = \mu(S_g) c_i$ that satisfy every group's first-order conditions and are consistent with $A$, since $\sum_i p_i^{1-\sigma} = \sum_g \mu(S_g)^{1-\sigma} K_g = A \sum_g S_g = A$. The left side of (13), $\mathcal{S}(A)$, is continuous and strictly decreasing. As $A \to \infty$ every share tends to zero. Let $\bar\psi = \lim_{S\to 1} \psi(S)$, infinite when $\eta = 1$; as $A$ falls toward $\max_g K_g/\bar\psi$ the largest group's share tends to one while every other share stays positive, so $\mathcal{S}$ exceeds one. Hence $\mathcal{S}(A) = 1$ has exactly one solution, and every other variable is a function of it. □

## B Entry

**B.1 Two elasticities.** Let $G(S) \equiv \Pi'(S)/\psi'(S)$, where $\Pi(S) = ES\,m(S)$, and define $\varepsilon_G = SG'/G$ and $\varepsilon_\psi = S\psi'/\psi$. Differentiating $\psi(S) = K/A$ gives $\partial S/\partial K = 1/(A\psi')$ and $\partial S/\partial A = -\psi/(A\psi')$. Hence $\Omega_K = G(S)/A > 0$, $\Omega_{KK} = G'(S)/(A^2\psi'(S))$, and

$$\Omega_{KA} = -\frac{1}{A^2}\Big[G'(S)\frac{\psi(S)}{\psi'(S)} + G(S)\Big] = -\frac{G(S)}{A^2 \varepsilon_\psi(S)}\big[\varepsilon_G(S) + \varepsilon_\psi(S)\big].$$

So $\Omega$ is strictly concave in $K$ if and only if $\varepsilon_G < 0$, and $\Omega_{KA} < 0$ if and only if $\varepsilon_G + \varepsilon_\psi > 0$.

**B.2 Proof of Theorem 1(a)–(b).** With $\eta = 1$, $\mu(S) = \sigma/[(\sigma-1)(1-S)]$. Set $E = 1$, $a \equiv 2(\sigma-1)$ and $b \equiv \sigma - 2 \geq 0$. Then $\Pi'(S) = (1 + aS)/\sigma$ and $\psi'(S) = C(1-S)^{-\sigma}(1+bS)$ with $C = [\sigma/(\sigma-1)]^{\sigma-1}$, so

$$\varepsilon_G(S) = S\,f(S), \qquad f(S) = \frac{a}{1+aS} - \frac{b}{1+bS} - \frac{\sigma}{1-S}.$$

At $S = 0$, $f(0) = a - b - \sigma = 0$. Since $a > b \geq 0$, $a/(1+aS) > b/(1+bS) \geq 0$, so $f'(S) = -a^2/(1+aS)^2 + b^2/(1+bS)^2 - \sigma/(1-S)^2 < 0$. Hence $f < 0$ and $\varepsilon_G < 0$ on $(0,1)$, which is (a). Next, $\varepsilon_\psi = (1+bS)/(1-S)$, and using $1 + bS - \sigma S = 1 - 2S$,

$$\varepsilon_G + \varepsilon_\psi = \frac{aS}{1+aS} - \frac{bS}{1+bS} + \frac{1-2S}{1-S}.$$

The map $x \mapsto x/(1+x)$ is increasing and $a > b$, so the first two terms sum to a positive number; the third is non-negative when $S \leq \frac{1}{2}$. This is (b). At $S = \frac{1}{2}$ the expression equals $(\sigma-1)/\sigma - (\sigma-2)/\sigma = 1/\sigma$. □

**B.3 Proof of Theorem 1(c).** *Step 1: the ladder.* Since $\Omega_K > 0$, a set $T$ is optimal only if no $T'$ has $K(T') \geq K(T)$ and $\Phi(T') \leq \Phi(T)$ with one inequality strict. The remaining efficient sets can be ordered so that $K$ and $\Phi$ rise together. *Step 2: single crossing.* For two rungs with $K_1 < K_2$, let $\Delta(A) = \int_{K_1}^{K_2} \Omega_K(K; A)\, dK - [\Phi_2 - \Phi_1]$ be the net gain from the larger one. Under the hypothesis, $\Omega_{KA} < 0$ on $[K_1, K_2]$, so $\Delta$ is strictly decreasing in $A$. Hence if the larger rung is preferred at some $A'$, it is preferred at every $A < A'$, and with no ties the optimal strength $K_g^*(A)$ is non-increasing in $A$ (Milgrom and Shannon, 1994). *Step 3: the aggregate.* Given strengths, the aggregate $A(\{K_g\})$ solving (13) is strictly increasing in each $K_g$: raising $K_j$ at a given $A$ raises the left side, which is strictly decreasing in $A$. *Step 4.* Suppose $(A, \{K_g\})$ and $(A', \{K_g'\})$ are equilibria with $A < A'$. By Step 2, $K_g' = K_g^*(A') \leq K_g^*(A) = K_g$ for all $g$, so by Step 3, $A' = A(\{K_g'\}) \leq A(\{K_g\}) = A$, a contradiction. Equilibria with the same aggregate have the same strengths, and with no ties the same active sets. □

**B.4 The notion of entry equilibrium and existence.** Definition 1 evaluates every alternative set at the prevailing aggregate $A$. In a two-stage game in which the market is re-solved after a deviation, a group that adds a plant anticipates that $A$ rises, which lowers the value of the deviation; the two notions coincide for groups with negligible shares and can differ for large ones. Theorem 1 applies to the notion of Definition 1. For the two-stage notion we have only numerical evidence, from markets small enough to enumerate every configuration and re-solve the market after every deviation. Existence in pure strategies is not guaranteed under either notion: a group's choice is discrete, so $K_g^*(A)$ jumps, and (13) may have no solution with each group on a rung. The algorithm of Section 1.3 then returns the configuration with the fewest groups wishing to deviate.

## C General Equilibrium

**C.1 Proof of Proposition 3.** *(a)* Let $T$ map log price indices $\ln P$ into the log price indices generated by the costs (4)–(6) that $P$ implies, holding wages and active sets fixed. Multiplying every cost in a market by a constant leaves shares and markups unchanged and multiplies every price by that constant, so each market's price index is homogeneous of degree one in the costs entering it; by Lemma 1's second step it is also non-decreasing in each of them. In (4), $\ln c_i$ depends on $\ln P$ through $\nu_k \sum_{k'} \omega_{k'k} \ln P_{\ell k'}$, with non-negative weights summing to $\nu_k$. Hence $T$ is monotone and $T(\ln P + x\mathbf{1}) \leq T(\ln P) + (\max_k \nu_k)\, x\mathbf{1}$ for $x \geq 0$, so by Blackwell's sufficient conditions $T$ is a contraction in the sup norm with modulus $\max_k \nu_k < 1$. *(b)* Given shares and markups, the second term of (16) is linear in market sizes, $E = \mathcal{E}X + BE$, where $B \geq 0$ and the column of $B$ for market $(m, k')$ sums to $\nu_{k'} \sum_i s_i/[\mu(1+t)] \leq \nu_{k'} < 1$. So $\rho(B) < 1$ and $E = (I - B)^{-1}\mathcal{E}X$. Variable profits and tariff revenue are linear in $E$, so (8) reads $X = wL - \theta'\Phi + \mathcal{M}X$ with $\mathcal{M} \geq 0$. Summing over countries, $\mathbf{1}'\mathcal{M}X$ equals world spending minus intermediate purchases minus the variable wage bill, that is world final spending minus the variable wage bill; since every unit of spending generates a

positive variable wage bill when $\nu_k < 1$, each column of $\mathcal{M}$ sums to less than one, $\rho(\mathcal{M}) < 1$, and $X = (I - \mathcal{M})^{-1}(wL - \theta'\Phi)$ is the unique solution. □

**C.2 Proof of Proposition 4.** Let $z_n(w) = W_n(w)/w_n - L_n$ be excess demand for labor. By Propositions 2–3, $z$ is continuous on the interior of the wage space, as a composition of continuous maps; it is homogeneous of degree zero; it satisfies Walras' law $\sum_n w_n z_n(w) = 0$, which follows from adding (8) across countries and using (16); and it is bounded below by $-L$. For the boundary condition, let $w_n \to 0$ with other wages bounded away from zero. Spending in every market stays bounded away from zero, since incomes are bounded below by $w_j L_j - \sum_g \theta_{gj}\Phi_g$, which we assume positive (fixed costs small relative to labor income). Delivered costs of plants located in $n$ fall at least in proportion to $w_n^{1-\nu_k}$, and their variable cost bills are $Es_i/[\mu(1+t)]$. If the group's share stays below one, the markup is bounded and the bill stays bounded away from zero. If the share tends to one (possible only when $\eta = 1$), (12) implies that $1 - S_g$ falls like $c_i^{(\sigma-1)/\sigma}$, so the bill falls like $w_n^{(1-\nu_k)(\sigma-1)/\sigma}$, more slowly than $w_n$. Either way $z_n = W_n/w_n - L_n \to \infty$. The standard existence result for excess demand functions (Mas-Colell, Whinston, and Green, 1995, Proposition 17.C.1) applies. □

**C.3 Proof of Lemma 1.** Hats denote log changes in response to a change in wages. *Costs.* From (4), $\hat{c}_i = (1-\nu_k)\hat{w}_\ell + \nu_k \sum_{k'} \omega_{k'k}\hat{P}_{\ell k'}$, with non-negative coefficients summing to one. *Markets.* Differentiating $S_g = \mu(S_g)^{1-\sigma}K_g/A$, with $\hat{K}_g = (1-\sigma)\hat{c}_g$ by (11) and $\hat{A} = (1-\sigma)\hat{P}$, gives $\hat{S}_g = -(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P})$. Shares sum to one, so $\sum_g S_g\hat{S}_g = 0$, which gives $\hat{P} = \sum_g \varpi_g \hat{c}_g$ with weights $\varpi_g > 0$ summing to one. *Stacking.* Substituting costs into markets gives $\hat{P} = \Gamma\hat{w} + \mathcal{N}\hat{P}$ with $\Gamma, \mathcal{N} \geq 0$, $(\Gamma + \mathcal{N})\mathbf{1} = \mathbf{1}$, and the rows of $\mathcal{N}$ summing to $\nu_k < 1$. Hence $\rho(\mathcal{N}) < 1$, $(I - \mathcal{N})^{-1} = \sum_{s\geq 0}\mathcal{N}^s \geq 0$, and $\partial \ln P/\partial \ln w = (I-\mathcal{N})^{-1}\Gamma \geq 0$ with row sums $(I-\mathcal{N})^{-1}(\mathbf{1}-\nu) = \mathbf{1}$. □

**C.4 A decomposition of factor payments.** Let $V_i$ be plant $i$'s variable cost in a market $(m,k)$, with tariffs fixed. Since $V_i = r_i/[\mu_g(1+t)]$, $\hat{V}_i = \hat{E} + \hat{s}_i - \hat{\mu}_g$, with $\hat{s}_i = (1-\sigma)(\hat{c}_i + \hat{\mu}_g - \hat{P})$ and $\hat{\mu}_g = \varepsilon_\mu \hat{S}_g$. Using C.3 and $(\sigma-1) = (\sigma-1)\Lambda_g[1 + (\sigma-1)\varepsilon_\mu]$,

$$\hat{V}_i = \hat{E} - (\sigma-1)(\hat{c}_i - \hat{c}_g) - (1-\varepsilon_\mu)(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P}). \tag{A.1}$$

The first term is the growth of the market, the second the reallocation of sales within the group, the third the reallocation between the group and its rivals. Its coefficient is non-negative if and only if $\varepsilon_\mu \leq 1$, which under quantity competition with $\eta = 1$, where $\varepsilon_\mu = S/(1-S)$, is exactly $S_g \leq \frac{1}{2}$.

**C.5 Proof of Proposition 5.** Raise $w_j$ alone, $j \neq n$. With $\nu_k = 0$ and head-office services a capability, the whole variable labor bill of a plant is paid in its location, so the plants paying country $n$ are located in $n$, and their costs do not respond: $\hat{c}_i = 0$. Fixed costs (7) are paid at $w_n$ and do not respond either. By Lemma 1, $\hat{c}_g, \hat{P} \in [0,1]$. Let $\chi = (1-\varepsilon_\mu)(\sigma-1)\Lambda_g$. The one-half condition and $\Lambda_g \leq 1$ give $0 \leq \chi \leq \sigma - 1$, so the last two terms of (A.1) equal $(\sigma-1)\hat{c}_g - \chi(\hat{c}_g - \hat{P})$. If $\hat{c}_g \geq \hat{P}$ this is at least $(\sigma-1)\hat{P}$; otherwise both parts are non-negative. In either case it is at least $(\sigma-1)\min\{\hat{c}_g, \hat{P}\} \geq 0$, so $\hat{V}_i \geq \hat{E}$. Weighting by $\pi^n_{i,m}$ and adding the fixed-cost part, whose response is zero, gives (18). □

**C.6 Gross substitutes and uniqueness.** The following classical argument is included for completeness. *If $M_{nj} > 0$ for $n \neq j$ at every point of a set $D$ that is convex in logarithms, $D$ contains*

*at most one equilibrium up to scale.* Let $w, w^* \in D$ be non-proportional equilibria, scale $w^*$ so that $w^* \le w$ with equality at some $n_0$, and move from $w^*$ to $w$ along the segment in logarithms. Every wage other than $w_{n_0}$ weakly rises and at least one strictly rises, so $W_{n_0}$ strictly rises while $w_{n_0}$ does not, and $z_{n_0}(w) > z_{n_0}(w^*) = 0$, a contradiction.

**C.7 Proof of Theorem 2.** In logarithms $\Psi$ becomes a map with Jacobian $(1-\lambda)I + \lambda M$, whose rows sum to one. Its off-diagonal entries are positive by (i) and its diagonal entries are positive by (ii), so it is a strictly positive stochastic matrix at every point of $D$. In logarithms $d_H$ is the oscillation seminorm $\operatorname{osc}(x) = \max_n x_n - \min_n x_n$. By the mean value theorem in integral form, the difference between the images of two points is $\bar{A}$ times the difference between the points, where $\bar{A}$ is the average of the Jacobian along the segment joining them, which stays in $D$. The contraction coefficient of a stochastic matrix in the oscillation seminorm, $\delta(A) = \frac{1}{2}\max_{i,j} \|A_{i\cdot} - A_{j\cdot}\|_1$, is convex in $A$, so $\delta(\bar{A}) \le \sup_D \delta$, and for a strictly positive matrix $\delta(A) \le \tanh(\Delta(A)/4) < 1$, where $\Delta(A) = \max_{i,j,k,l} \ln(A_{ik}A_{jl}/A_{jk}A_{il})$ (Birkhoff, 1957). Taking $\tau = \sup_D \tanh(\Delta/4)$ gives the contraction. Two fixed points in $D$ would contradict it; convergence additionally needs the iterates to stay in $D$. □

**C.8 Head-office services as a factor input.** Suppose instead that the labor term in (4) is $(w_h^{\alpha_k} w_\ell^{1-\alpha_k})^{1-\nu_k}$, so that country $h$ is paid the share $\alpha_k$ of the labor bill of every plant its groups run. All plants of a group report to the same headquarters, so country $h$ receives the share $\alpha_k$ of the group's whole bill in the market, and the within-group term of (A.1) averages to zero over the group. Consider a country $n$ whose labor income consists only of head-office payments from groups whose plants are all located in a single country $j$, with $\nu_k = 0$. Raising $w_j$ raises those groups' costs by $\hat{c}_g = 1 - \alpha_k$, while the price indices of the markets they serve rise only in proportion to the groups' incidence weights. By (A.1), $M_{nj}$ is then a weighted sum over these groups of $\hat{E} - (1-\varepsilon_\mu)(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P})$. When the groups are small, $\Lambda_g \approx 1$, $\varepsilon_\mu \approx 0$ and $\hat{c}_g - \hat{P} \approx 1 - \alpha_k$, so $M_{nj}$ is negative whenever $(\sigma-1)(1-\alpha_k)$ exceeds the growth of the markets they serve, although no group is near one half of any market. The failure is worst when $\alpha_k$ is small: a large head-office share ties the group's cost to its own country's wage and insulates it from $w_j$. What breaks gross substitutes is a country whose income is concentrated on firms whose costs are set by a single foreign country.

## D Comparative Statics

**D.1 Proof of Proposition 6.** The first two equations are the market step of C.3, with the active sets and $E$ fixed. If only $j$'s costs change, $d\ln P = \varpi_j \hat{c}_j$, so $d\ln S_j = -(\sigma-1)\Lambda_j(1-\varpi_j)\hat{c}_j$ and, for $h \ne j$, $d\ln S_h = (\sigma-1)\Lambda_h \varpi_j \hat{c}_j$. Prices are $p_i = \mu(S_g)c_i$, so $d\ln p_i = \hat{c}_i + \varepsilon_\mu(S_g)\, d\ln S_g$, which gives $\rho_j$ for group $j$ and a positive price change for every rival. Since $(\sigma-1)\varepsilon_\mu\Lambda_j < 1$ and $1-\varpi_j \in [0,1)$, $\rho_j \in (0,1]$. Under quantity competition with $\eta = 1$, $\varepsilon_\mu = S/(1-S)$, which gives the stated $\Lambda_g$. □

**D.2 Proof of Proposition 7.** With $m(S) = (1+\kappa S)/\sigma$, $\Pi^H/E = \sigma^{-1}\sum_g \theta_g S_g(1+\kappa S_g) = \sigma^{-1}[\bar{\theta} + \kappa \sum_g \theta_g S_g^2]$, and $\sum_g \theta_g S_g^2 = \bar{\theta}\cdot \mathrm{HHI} + \mathrm{Cov}_S(\theta, S)$ by definition. Total profit is the case $\theta_g \equiv 1$, $\sum_g \Pi_g/E = (1+\kappa\,\mathrm{HHI})/\sigma$, using $\sum_g S_g = 1$. Subtracting $\bar{\theta}\sum_g \Pi_g$ from $\Pi^H$ leaves $(E\kappa/\sigma)\mathrm{Cov}_S(\theta, S)$. Corollary 1 is the case $\kappa = 0$. □

**D.3 Proof of Proposition 8.** By (14), $d\Pi_g = (E/\sigma)(1+2\kappa S_g)\, dS_g$. Let $Z = \sum_h S_h\Lambda_h$, so $\varpi_j = S_j\Lambda_j/Z$. From D.1, $dS_j = -(\sigma-1)S_j\Lambda_j(1-\varpi_j)\hat{c}_j = -(\sigma-1)\varpi_j\hat{c}_j \sum_{h\ne j} S_h\Lambda_h$ and $dS_h =$

$(\sigma-1)\varpi_j S_h \Lambda_h \hat{c}_j$ for $h \neq j$. Substituting into $d\Pi^H = \sum_g \theta_g d\Pi_g$ and collecting terms gives (22). □

**D.4 Proof of Corollary 2.** Monotonicity in $A$ is Step 2 of B.3. For spending, $\Omega(K;A) = E\,\tilde{\Omega}(K;A)$ with $\tilde{\Omega}$ strictly increasing in $K$, so for two rungs with $K_1 < K_2$ the net gain $E[\tilde{\Omega}(K_2;A) - \tilde{\Omega}(K_1;A)] - [\Phi_2 - \Phi_1]$ is strictly increasing in $E$, and the larger rung, once preferred, remains preferred as $E$ rises. □